

# Relation Algebra

## §6. Maps

(6a)

|                                                                                      |                                                            |
|--------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>surjective</b><br>$\mathbb{1} \sqsubseteq R^\circ R$ , that is, $R^\circ$ entire. | <b>single valued</b><br>$R^\circ R \sqsubseteq \mathbb{1}$ |
| <b>entire</b><br>$\mathbb{1} \sqsubseteq RR^\circ$ . With single valued, a map.      | <b>injective</b><br>$RR^\circ \sqsubseteq \mathbb{1}$      |

(6b)

| law                                                                         | picture                                                                |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------|
| $\text{Simple}(F) \implies F(R \cap S) = F R \cap F S$<br>$F$ single valued | <p>one person who admires both</p> <p><b>a</b> at A, <b>b</b> at B</p> |